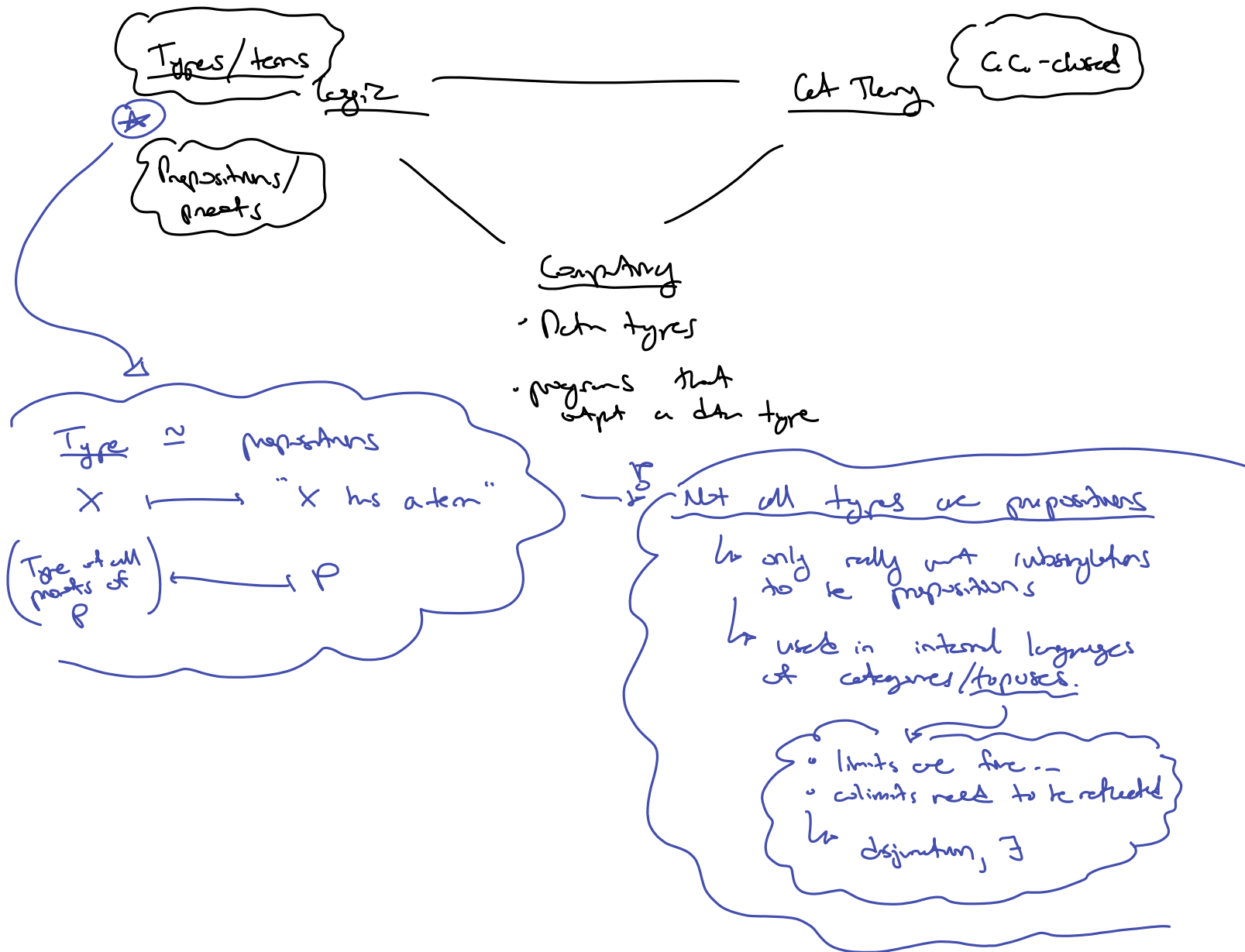


Curry-Howard-Lambek



How do we define types?

Logically:

Formation —→ "there is a type..."

Introduction —→ "what are the terms..."

Elimination —→ "how do we access these terms..."

Categorically

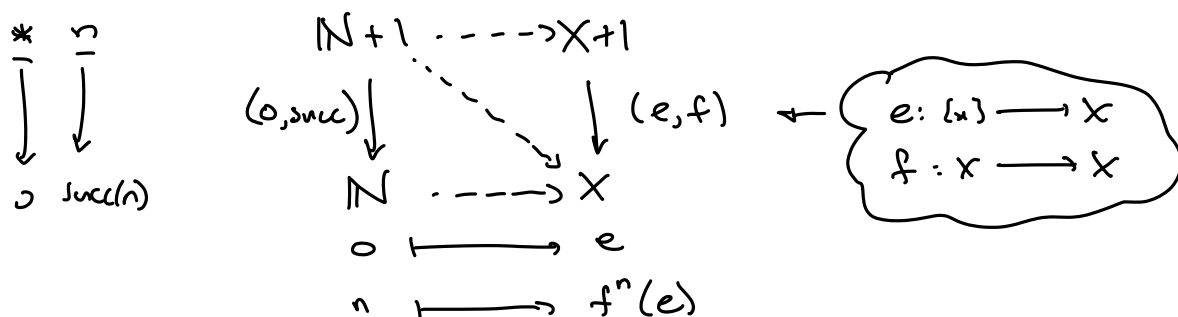
"Universal Property"

Some of these types are "inductive" and their elimination rule can be called "induction"

INDUCTION: categorically, this is a universal property coming from an initial algebra for an endofunctor on $\mathcal{U}$ or $\mathcal{V}$.

Example: $\text{Set} \xrightarrow{+1} \text{Set}$
 $X \longmapsto X \sqcup \{*\} = X+1$

and $\mathbb{N}$ are an initial algebra for this endofunctor..



Example: $A: \mathcal{U} \rightarrow \mathcal{U}$ (a category), $\mathcal{U}_{A \times A} \xrightarrow{!} \mathcal{U}_{A \times A}$
 $\begin{pmatrix} E \\ \downarrow \\ A \times A \end{pmatrix} \longmapsto \begin{pmatrix} A \\ \downarrow \Delta \\ A \times A \end{pmatrix}$

What is an initial algebra for $!$? and what does induction mean? ... we will revisit...

Equality (for terms of a given type)

Definitionally

egs $f(x) = x+1$
then $f(1) = 1+1$, by def'n
 $" \equiv "$, $" := "$

Propositionally

egs $x^2 - 1 = (x-1)(x+1)$

This is something I need to prove, by expanding, cancelling, etc...

Classically, these are assumed to be substitution types ...
So if $a=b$, then it happens

is not one way,
In HoTT this gets relaxed...

Defining equality/identity types

Formation: $\frac{A: \mathcal{U}}{a, b: A}$
 $a =_A b : \mathcal{U}$

Introduction: $\text{refl}: \prod_{a:A} (a =_A a)$

The easiest way to see that $a=b$, is if they are the same by definition

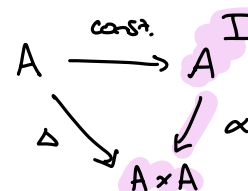
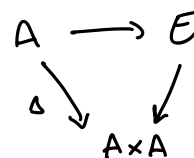
Elimination: (Path Induction)

$C: \prod_{x,y:A} (x =_A y) \longrightarrow \mathcal{U}$, $c: \prod_{x:A} C(x, x, \text{refl}_x)$

$f: \prod_{x,y:A} \prod_{p:(x=y)} C(x, y, p)$ such that $f(x, x, \text{refl}_x) = c(x)$

Going back ... A Δ -algebra is a morphism
leads to $\mathcal{U}/_{A \times A} \xrightarrow[\text{pb } \Delta]{\text{Eq}} \mathcal{U}/_A$ Heyting sees this

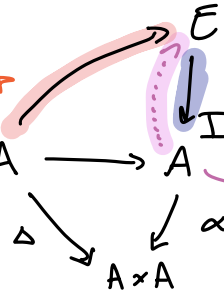
The initial algebra is the path space object



An Δ -algebra over the initial algebra

is a diagram
section of the pb of E over A
interpret as substitution

$E(a, a, \text{refl}_a)$



interprets a type

$a, b: A, p: (a=b) \vdash E(a, b, p)$

The elimination rule, says that we can extend the section to a section A^I .

"If it holds for all refl_a , then it holds for all paths..."

↳ "A proof of E that holds for all reflexive id's ($\text{refl}_a: a=a$) extends to a proof of E that holds for all id's ($p: a=b$)."

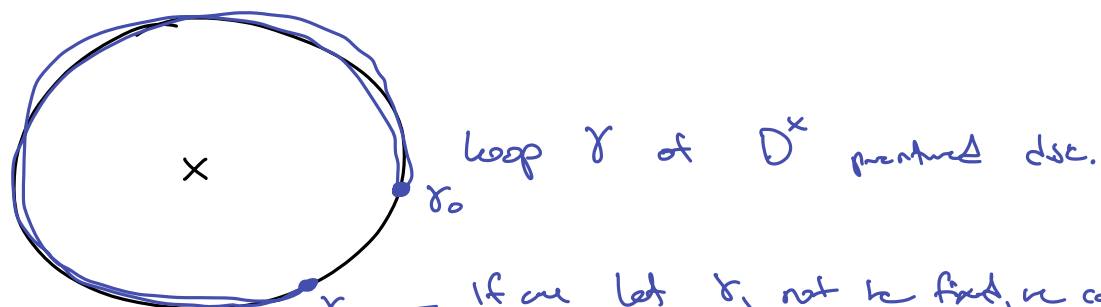
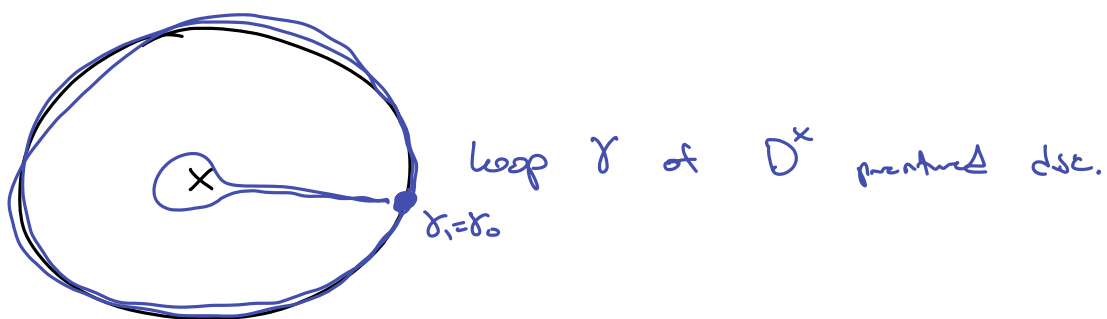
TRANSLATION

<u>II</u>	<u>H₀</u>	<u>Coat</u>
Types	Spaces	∞ Grpd
terms	points	objects
identity types	path spaces	$\text{Hom}(-, -)$
refl	constant path	Id-morphism.
$\vdots$	$\vdots$	$\vdots$

Note: the identity type is not inductive, but the identity family is.

↳ Homotopically: the based path space is contractible, we can homotope our paths back to the constant loop...
 ↳ (not all paths are constant...)

But: path induction does not give us an element of a fam $C(p)$, if path p has 2 fixed endpoints.



— if we let x_1 not be fixed, we can retract the path back.

Lemma: $\forall A:U, x, y:A$, there's a function

$$(x =_A y) \xrightarrow{(-)^{-1}} (y =_A x)$$

such that $\text{refl}_x^{-1} = \text{refl}_x$ for all $x:A$.

This is a proposition, so it's a type.

this is a separate judgement.

$$\prod_{A:U} \prod_{x, y:A} (x =_A y) \longrightarrow (y =_A x)$$

and a proof is a term of this type.

Pf: (w/ path induction) Given $A:U$, define a family

$$\bullet D: \prod_{x, y:A} (x =_A y) \longrightarrow U \text{ by } D(x, y, p) = (y =_A x)$$

and a function

$$\bullet d: \prod_{x:A} D(x, x, \text{refl}_x) \text{ by } \underbrace{\lambda x. \text{refl}_x}_{d} : (x =_A x)$$

• The induction principle extends d to a function

$$\underbrace{\lambda p. \text{ind}_{=A}(D, d, x, y, p)}_{\text{call this } (-)^{-1}} : (y =_A x)$$

$$\text{that } p^{-1} := \text{ind}_{=A}(D, d, x, y, p)$$

$$\text{notice: } x:A, \text{refl}_x^{-1} := \text{ind}_{=A}(D, d, x, x, \text{refl}_x)$$

$$= d(x)$$

$$= \text{refl}_x \quad /$$

Pf: (informal) Given $A: \mathcal{U}$, $x, y: A$, $p: x =_A y$,
 we want to define $\bar{p}: (y =_A x)$.

By path induction, it suffices to consider when x and y
 are the same point and p is the constant path.

Then we can just define

$$\text{refl}_x^{-1} = \text{refl}_x : (x =_A x) \text{ as the "base case"}$$

The general case follows by (path) induction,
 and the "conversion rule" (what can alter "such that")
 is precisely the base case. //

Let's prove things to prove similarly:

* path concatenation

$$\begin{array}{ccccc} (x =_A y) & \longrightarrow & (y =_A z) & \longrightarrow & (x =_A z) \\ p & \longmapsto & q & \longmapsto & p * q \end{array}$$

TRANSLATION

IT
 reflexivity
 symmetry
 transitivity

 ' = ' is an
 equivalence rel'n

Ho
 const. path
 inverse path
 path concatenation

Cat
 Id-morphism
 inverse-morphism
 morphism composition

 this is an eqpd

Lemma: (Path Concat) $\forall A: \mathcal{U}, x, y, z: A,$

$$\begin{array}{ccccc} (x \underset{A}{=} y) & \longrightarrow & (y \underset{A}{=} z) & \longrightarrow & (x \underset{A}{=} z) \\ p & \longmapsto & q & \longmapsto & p \circ q \end{array}$$

Rewrite as a type:

$$\prod_{A: \mathcal{U}} \prod_{x, y, z: A} (x = y) \longrightarrow (y = z) \longrightarrow (x = z)$$

Define a family: $D: \prod_{x, y: A} (x = y) \longrightarrow \mathcal{U}$

$$D(x, y, p) = \left(\prod_{z: A} (y = z) \longrightarrow (x = z) \right)$$

Define $d: \prod_{x: A} D(x, x, \text{refl}_x)$ to be

$$\lambda x \left(\lambda z \ p. p \right) \quad \left\{ \prod_{z: A} (x = z) \longrightarrow (x = z) \right\}$$

d

$$d(x) : \prod_{z: A} (x = z) \longrightarrow (x = z)$$

$$d(x) : \prod_{z: A} (x = z) \longrightarrow (x = z)$$

$$(d(x)(z))(p) = p$$

By induction, we extend $d \dots \ddot{}$

More things we can prove similarly

* assoc of path concat. (up to htpy...)

* unit laws

* inverse laws

* $(-)^{-1}$ is idempotent

leave for Jeremiah/laurens.

$\hookrightarrow (\text{refl refl refl} \dots)$

These are all propositional equalities above

eg $p * (q * r) = (p * q) * r$

Is an identity type of identity types

a path between paths

i.e. a (higher) homotopy!

Technically, we need more coherence laws as we go higher,
but all of this is generated from the path induction
for id-types.

Loops

$A : \mathcal{U}$, $a : A$, define $\Omega(A, a) := (a =_A a)$

• Path concatenation gives an operation

$$\Omega(A) \times \Omega(A) \xrightarrow{*} \Omega(A)$$

and the (higher) groupoid structure of $(a =_A a)$ gives it
the structure of a "higher group"

• Higher loops! $\Omega^2(A) = \Omega(\Omega(A, a), \text{refl}_a)$

this is also a "higher group", but it has 2 operations
and by E.H. it's abelian.

THM: $\Omega^2(A) \times \Omega^2(A) \xrightarrow{*} \Omega^2(A)$ is commutative.

Really, start with the type $(A, a) \rightsquigarrow A : \mathcal{U}, a : A$.

and then $\mathcal{U}_\bullet = \sum_{A : \mathcal{U}} A$

Then $\Omega(A, a) = (\lambda a \in A, \text{refl}_a)$

And $\Omega^n(A, a)$ recursively,

$$\Omega^0(A, a) = (A, a)$$

$$\Omega^{n+1}(A, a) = \Omega^n(\Omega(A, a))$$